Q3C: Evidence-Conditioned Candidate Confidence for Safer Abstention in Open-Domain Question Answering

Anonymous ACL submission

Abstract

Small open-weight language models are attractive for open-domain question answering because they are cheap to adapt and serve, yet they routinely return wrong factual answers with high apparent confidence, and common signals such as token probability, verbal uncertainty, or retrieval presence do not reliably indicate whether a candidate answer is supported. We propose **Q3C**, a QLoRA-adapted question–candidate–context judge that scores each answer candidate from the question, available evidence, sampled alternatives, and benchmark metadata, then answers only when the selected candidate clears a predeclared high-confidence operating point. The central idea is to move confidence estimation from the generated answer to the *answer candidate in context*, so that evidence support, candidate semantics, and cross-sample agreement are judged jointly rather than collapsed into a single scalar. On a 300-task paired evaluation spanning TriviaQA, Natural Questions, and TruthfulQA with 14 methods and ablations under a 7B-class open-model backbone, Q3C raises raw score over direct raw confidence from 22.00% to 31.67% and reduces overconfident-error rate from 76.00% to 12.00%, with evidence-conditioned ablations showing that support context contributes to the trade-off. The improvement is bounded: retrieval-only RAG remains non-dominated with 0.00% overconfident-error rate, and Q3C's current abstention gate collapses TruthfulQA effective utility despite nonzero raw correctness. These results support candidate-level, evidence-conditioned confidence as a useful reliability layer for small-model open-domain QA, while showing that operating-point design is central to safer abstention.

1 Introduction

Open-domain question answering exposes a reliability problem that is sharper than ordinary accuracy loss. A small instruction-tuned language model can return a plausible factual answer, attach high verbal or self-reported confidence to it, and still be wrong. In user-facing QA this is a *confident hallucination*: the answer is not merely unsupported, but is presented with enough confidence to discourage checking. The problem is especially important for 7B–8B open models, which are attractive for deployment because they are affordable to adapt and serve, but have less factual capacity than frontier closed systems.

This paper focuses on the deployment regime where a small open model is expected to answer many short factual questions rather than write long essays. In that regime, a single wrong answer can be cheap to generate but costly for the user to detect. Recent work on small-model and TruthfulQA-style hallucination diagnostics shows that confident mistakes often concentrate around named entities, negation, numerical facts, and other symbolic triggers (Lamba et al., 2025b,a; Adeseye et al., 2026). Work on verbal uncertainty further shows that asking a model to state uncertainty can reduce some hallucination symptoms, but only if the uncertainty signal is calibrated to correctness rather than treated as a polite hedge (Ji et al., 2025). We therefore evaluate confidence as an operational decision variable: a high-confidence wrong answer is counted separately from an ordinary wrong answer, and abstention is scored separately from raw correctness.

Calibration and selective prediction provide a starting point. Classical calibration shows that neural confidence can be systematically misaligned with correctness (Guo et al., 2017), while selective prediction asks the system to abstain when the risk of answering is too high. Free-form QA complicates both ideas: correctness is semantic rather than class-indexed, the candidate answer may be one of many aliases, and probability mass over tokens is not a direct probability that the proposition is true. Recent analyses of hallucination and selective prediction further suggest that memory conflict, commitment to a wrong answer, and entropy-only abstention can leave high-confidence errors unresolved (Liang et al., 2026; Yeom et al., 2026; Phillips et al., 2026).

The same tension appears in broader uncertainty-estimation work. Surveys and benchmarks for LLM uncertainty emphasize that token-level entropy, verbalized probabilities, semantic clustering, and learned probes each capture only part of the reliability problem (Xia et al., 2025; Tao et al., 2025; Sung et al., 2025). One-pass distillation, concept-level uncertainty, and graph-enhanced uncertainty methods reduce the cost of repeated sampling or add structure to the signal (Zhao et al., 2025; Wang et al., 2024, 2025), while truth-aligned uncertainty work argues that confidence should be judged against factual correctness rather than fluency (Srey et al., 2026). These studies motivate our design choice: the confidence module should see the answer candidate, evidence context, and alternatives together, because no single scalar from the generator is expected to be sufficient.

Sampling and retrieval address different pieces of the problem. SelfCheckGPT and semantic-uncertainty methods compare multiple generations to detect unstable answers (Manakul et al., 2023; Kuhn et al., 2023; Aichberger et al., 2024). Retrieval-augmented generation and self-reflective retrieval can expose evidence and improve factuality (Lewis et al., 2020; Asai et al., 2023). Yet agreement among samples is not evidence support, and the presence of a passage does not guarantee that the passage supports the answer; partial evidence can also anchor a wrong completion (Lathkar, 2026). The central gap is therefore not whether the model can produce another answer or fetch text, but whether it can judge a *candidate answer* in context.

Retrieval-specific evidence also argues against treating RAG as a solved calibration layer. Recent RAG analyses report hallucination and coverage errors even when retrieval is present, uncertainty-sensitive retrieval policies that must decide whether to retrieve at all, and mechanistic or graph-consistency detectors for retrieved evidence (Chang et al., 2024; Dhole, 2025; Sun et al., 2024; Shen, 2026). Multi-source RAG work similarly stresses that evidence aggregation is itself a modeling problem rather than a guarantee of truth (Wu et al., 2025). Our experiments therefore include retrieval-only and retrieve-and-critique baselines, but the proposed method is not a replacement for retrieval. It is a candidate-confidence layer that can be evaluated against retrieval as a nontrivial competitor.

We propose **Q3C**, a question-candidate-context confidence judge. Q3C adapts a small open instruction model with QLoRA supervision to read the question, a candidate answer, evidence snippets when available, sampled alternatives, and benchmark metadata. It predicts support and confidence for candidate correctness, ranks candidates, and abstains when the selected candidate falls below the operating threshold. Figure 1 gives the paper-facing overview. Unlike prompt-only self-critique, Q3C is trained as a candidate-confidence module; unlike retrieval-only confidence, it separates evidence availability from answer support; unlike an oracle policy, it is evaluated on held-out benchmark rows with raw correctness retained separately from abstention-adjusted utility.

Our evidence is a completed 4200-row matrix over 14 methods and ablations on TriviaQA (Joshi et al., 2017), Natural Questions (Kwiatkowski et al., 2019), and TruthfulQA (Lin et al., 2021). These benchmarks are deliberately complementary: factoid aliases, naturally occurring user questions, and misconception-sensitive truthfulness stress different failure modes. Q3C improves raw score over direct raw confidence by 9.67 points and lowers overconfident-error rate by 64.00 points under paired task evaluation. The same matrix also limits the claim: retrieval-only RAG remains a strong low-OCER baseline, the fixed-threshold ablation matches full Q3C on aggregate metrics, and TruthfulQA reveals an abstention failure in which raw-correct answers are rejected. The contribution is therefore a calibrated method and analysis, not a universal win claim.

2 Related Work

Calibration and selective prediction. Modern neural models can be poorly calibrated even when their accuracy is useful, motivating post-hoc calibration such as temperature scaling (Guo et al., 2017). Selective prediction extends this goal by allowing abstention when the model estimates high risk. LLM-specific selective-prediction work studies how to align language models with answer/refuse behavior, while self-evaluation adaptation trains models to judge their own outputs (Luo et al., 2026; Chen et al., 2023). Q3C differs by making the judged object an answer candidate paired with evidence and alternatives, rather than only a generated response and a scalar confidence.

Recent selective-prediction variants sharpen that distinction. Density-ridge selective prediction studies label-scarce hallucination detection and warns that supervised probes can degrade away from their calibration labels (Shamsi, 2026). Memory-augmented selective prediction adds data to support refusal decisions (Sarkar et al., 2026), while work on selective "selective prediction" studies unnecessary abstention as a failure of its own (Srinivasan et al., 2024). Q3C inherits this risk: a low overconfident-error rate is not enough if the model refuses answerable items. We therefore report raw score, effective score after abstention, coverage, and OCER together rather than optimizing one number.

Sampling and semantic uncertainty. SelfCheckGPT detects hallucinations by measuring consistency across sampled generations without an external database (Manakul et al., 2023). Semantic uncertainty and semantically diverse generation improve on string-level disagreement by grouping meaning-equivalent answers (Kuhn et al., 2023; Aichberger et al., 2024). Q3C includes sampling-based baselines and ablations because these signals are strong competitors. Its premise is narrower: semantic agreement is useful, but a cluster of agreeing answers may still encode the same unsupported misconception.

Decoding strategy matters for these baselines. If diversity is too low, self-consistency collapses to a restatement of the greedy answer; if it is too high, candidate clusters can be dominated by irrelevant alternatives. Recent decoding and uncertainty work shows that the sampling procedure itself changes the uncertainty estimate (Hashimoto et al., 2025). Q3C uses sampled alternatives as features rather than as the final decision rule, which lets the learned judge treat agreement as one signal among evidence support, candidate semantics, and benchmark metadata.

Retrieval and evidence support. Retrieval-augmented generation supplies non-parametric evidence for knowledge-intensive tasks (Lewis et al., 2020), and Self-RAG-style methods train or prompt models to retrieve and critique their generations (Asai et al., 2023). The Q3C matrix confirms that retrieval is a serious baseline: the retrieval-only condition attains zero overconfident-error rate in our run. However, retrieval confidence is not identical to candidate support, especially when evidence is partial or misleading (Lathkar, 2026). Q3C therefore reports retrieval as non-dominated rather than treating it as a weak control.

Open-domain QA and truthfulness. TriviaQA, Natural Questions, and TruthfulQA stress complementary aspects of factual QA: alias-heavy trivia evidence, naturally occurring user questions, and misconception-sensitive truthfulness (Joshi et al., 2017; Kwiatkowski et al., 2019; Lin et al., 2021). We use them as three independent benchmark families. This matters because a confidence rule that works on factoid aliases can still fail on truthfulness: our failure analysis shows precisely that Q3C's current abstention policy removes all raw-correct TruthfulQA answers from effective credit.

Open-domain QA evaluation has itself changed under instruction-tuned LLMs. Evaluations of LLM-era ODQA argue that generated answers should be judged with attention to aliases, answerability, and retrieval context rather than by exact string match alone (Kamalloo et al., 2023). Natural Questions baselines and follow-up systems show that answerability, passage choice, and short-answer extraction are separable sources of error (Alberti et al., 2019; Pan et al., 2019). These observations inform our metric split: token-level F1 is useful where aliases matter, but OCER and abstention-adjusted utility are needed to measure whether confidence behavior is safer.

3 Problem and Q3C Method

Let q be a question, E_q the evidence context available to the evaluation condition, and $\mathcal{A}_q = \{a_0, \ldots, a_k\}$ the greedy and sampled answer candidates generated by the base model. A method outputs an answer a, a confidence $c \in [0, 1]$, and optionally an abstention decision. We count an overconfident error when the benchmark scorer marks the answer wrong and $c \geq 0.75$, the predeclared high-confidence threshold used in the analysis artifacts. We report both raw benchmark correctness and abstention-adjusted effective score so that a policy cannot look safe merely by refusing answerable questions.

Decision objective. Q3C is trained and evaluated as a *candidate selector with confidence*, not as a new QA generator. The generator first proposes candidates, and the confidence judge chooses whether a candidate should be answered. This separation is important for two reasons. First, direct generation can be wrong because the model never considers the right answer, but it can also be wrong because it commits to the wrong candidate despite semantically plausible alternatives (Yeom et al., 2026). Second, retrieval can provide a passage that is relevant to the question but does not support the selected answer (Lathkar, 2026). Q3C's input is therefore arranged around the pair (q, a_i): the question, one candidate answer, evidence snippets when available, other sampled candidates, and a compact metadata field naming the benchmark family and scoring affordances.

Supervision and decontamination. The QLoRA training examples are candidate-level judgments derived from benchmark training or development rows that are separated from the final 300-task matrix by task identifier. Each example asks the judge to assign a support state: supported, contradicted, unsupported, ambiguous, or abstain. The target confidence bin is tied to the candidate's judged correctness under the benchmark scorer, not to the base model's self-reported confidence. This makes the learned target closer to selective-prediction correctness probes than to free-form self-critique (Chen et al., 2023; Phillips et al., 2026), but the input structure is QA-specific: evidence snippets and alternatives are explicit fields rather than hidden prompt context.

Candidate construction. All compared conditions use `Qwen/Qwen2.5-7B-Instruct` through vLLM. Direct baselines use the greedy answer. Sampling baselines form answer clusters from six sampled candidates. Retrieval conditions receive benchmark-linked or task-provided evidence when the benchmark supports it. Q3C consumes the candidate answer, alternatives, evidence text, and metadata describing whether the benchmark has alias sets, evidence snippets, or truthfulness labels.

Confidence-judge adaptation. Q3C is trained as a QLoRA SFT adapter with LLaMA-Factory rather than as a hand-written rule. The supervised target asks the model to judge whether a candidate is supported, contradicted, unsupported, ambiguous, or should be abstained, and to emit a confidence bin used by the selection policy. The input deliberately combines three signal families: candidate semantics, evidence-answer agreement, and alternative-answer consistency. The no-evidence, answer-only, no-alternatives, and no-semantic-feature ablations remove these families to test whether the full input is doing more than changing prompt format.

Why candidate confidence rather than answer confidence? The method is motivated by the failure of answer-level confidence to distinguish three cases that look similar in a generated response. In the first case, all candidates agree and the evidence supports the answer; high confidence is appropriate. In the second, candidates agree because they share the same misconception; sampling confidence is high but support is absent. In the third, evidence is present but only partially anchors the reasoning chain; retrieval confidence is high but the answer remains unsupported. Q3C's input format is meant to expose these distinctions to the judge. The ablations are intentionally aligned with this design: removing evidence tests the second/third distinction, removing alternatives tests whether candidate diversity matters, and removing semantic features tests whether answer equivalence information is doing more than lexical matching.

Selection and calibration. At inference time, Q3C ranks candidates by supported-correctness confidence. The evaluated operating point answers only when the selected candidate receives the high-confidence bin, calibrated to 0.90 in the completed rows; medium and low bins, calibrated to 0.45 and 0.10, abstain. We also report a no-abstention variant and a post-hoc threshold sweep over completed rows. The sweep is analysis only: it changes accept/reject decisions over existing rows and is not claimed as a new trained method.

Accounting for abstention. An abstention policy can reduce visible errors by refusing nearly everything, which would be misleading for open-domain QA. For this reason, every Q3C-family result reports raw benchmark score and effective score after abstention. Raw score answers the question "did the selected candidate match the benchmark answer?" Effective score answers "did the system deliver a correct answer to the user?" OCER answers a third question: among all tasks, how often did the system produce a wrong answer at high confidence? The three measurements are not interchangeable, and the TruthfulQA failure in Section 7 depends on keeping them separate.

4 Experimental Setup

Evaluated system specification. The evaluated backbone is Qwen/Qwen2.5-7B-Instruct. Q3C and the generic self-evaluation baseline use LoRA adapters trained with LLaMA-Factory; inference and scoring use vLLM. The completed evaluation matrix contains 4200 rows: 14 methods or ablations, 300 paired tasks per condition, and 100 tasks from each benchmark family. The supplementary package stores raw scored rows, manifests, and generated analysis tables so that every number in the paper can be regenerated from completed rows rather than from the PDF.

Setup at a glance. The reader-facing evaluation is: *sources*–300 paired tasks, with 100 each from TriviaQA unfiltered-web development examples, Natural Questions development/validation examples, and TruthfulQA public CSV examples; *model/backend*–Qwen2.5-7B-Instruct served by vLLM, with LoRA judge adapters for Q3C and B6; *roles*–B0 direct confidence, B1 temperature scaling, B2 self-consistency, B3 semantic entropy, B4 retrieval-only RAG, B5 retrieve-and-critique, B6 generic self-evaluation, Q3C, and Q3C ablations; *Q3C mechanism*–rank answer candidates from the question, candidate answer, evidence when available, six sampled alternatives, and benchmark metadata, then answer only for the high-confidence bin; *budget/metrics*–one greedy primary answer for direct/retrieval/critique/learned-judge conditions, six sampled candidates for sampling and Q3C alternatives, greedy adapter judging with a 96-token cap, and raw score, effective score, F1, confidence, ECE, abstention, and OCER. The compact takeaway is that Q3C changes B0 raw score from 22.00% to 31.67% and OCER from 76.00% to 12.00%, while B4's 0.00% OCER and Q3C's TruthfulQA abstention failure bound the claim.

Benchmarks. The 300-task matrix draws 100 paired rows from each benchmark family: TriviaQA unfiltered-web development examples with alias-aware scoring, Natural Questions development/validation examples with answerability signals, and TruthfulQA public CSV rows with correct and incorrect answer fields. The matrix reports raw score, token-level F1 where applicable, mean confidence, ECE, abstention rate, and OCER; Q3C-style methods additionally separate raw selected-candidate correctness from effective score after abstention.

Task sampling and pairing. The same task IDs are used for every method and ablation. Pairing prevents gains from reflecting an easier family slice and supports McNemar tests over task-level binary outcomes rather than independent aggregate proportions.

Baselines and ablations. B0 is direct generation with raw self-reported confidence. B1 applies temperature scaling. B2 uses self-

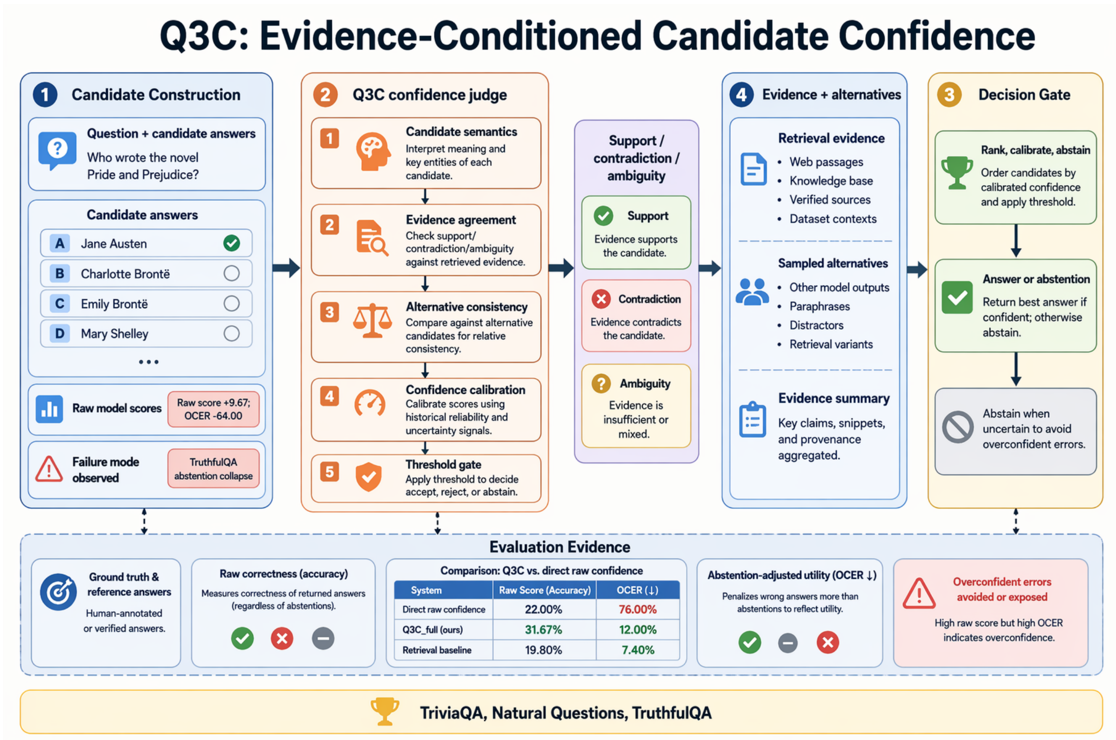

Figure 1: Q3C conceptual overview: the judge combines question/candidate inputs, evidence and alternatives, support judgment, calibration, and an answer-or-abstain gate. The evaluated matrix shows a +9.67 point raw-score gain and a -64.00 point OCER change versus direct raw confidence, while the failure path marks the TruthfulQA abstention collapse.

consistency over sampled candidates. B3 uses semantic-entropy confidence. B4 is retrieval-only RAG confidence. B5 is a Self-RAG-style retrieve-and-critique prompt. B6 is a generic learned self-evaluation adapter that does not receive Q3C's evidence-candidate feature structure. Q3C ablations remove evidence, alternatives, semantic features, abstention, or replace the learned operating point with the fixed threshold.

Decoding and budget policy. Direct, temperature-scaled, retrieval-only, retrieve-and-critique, and learned-judge conditions use one greedy primary answer per task. Sampling and semantic-entropy baselines use six sampled candidates; Q3C receives the same primary candidate, the six alternatives, and any evidence fields available to the matched condition. Adapter judging is greedy with a 96-token cap. Abstention is applied only after candidate scoring, so raw score measures selected-candidate correctness before refusal and effective score measures delivered answers after refusal.

Metrics. Raw score is exact-match or task-specific correctness after benchmark normalization, and F1 is reported where token-overlap aliases are meaningful. ECE bins predicted confidence against empirical correctness; OCER is the percentage of all tasks that are wrong with confidence at least 0.75. Abstention rate and effective score distinguish refusing from delivering a correct answer, so ECE, OCER, and effective score are reported together.

Statistical tests and scope. Main binary comparisons use exact McNemar tests over paired task outcomes, with Wilson intervals in the generated analysis artifacts. These tests support claims about this 300-task matrix, not all seeds or all small language models; the threshold sweep is post-hoc analysis over completed Q3C rows rather than a deployed policy.

5 Main Results

Table 1 is the central paper matrix. Q3C improves raw benchmark score over B0 direct raw confidence from 22.00% to 31.67% with a paired McNemar $p = 0.00002490$, and lowers OCER from 76.00% to 12.00% with $p < 10^{-8}$. The same table shows why the claim is metric-specific: B4 retrieval-only RAG has 0.00% OCER and nearly identical mean F1, while Appendix Table 3 shows that Q3C's abstention-adjusted utility trails its no-abstention variant.

The confidence columns explain the mechanism behind the headline. B0 is wrong on many rows and still reports high confidence, yielding a mean confidence of 0.899 and OCER of 76.00%. Temperature scaling reduces OCER to 7.33% but does not improve raw score, which is expected for a post-hoc scalar calibration method. Q3C changes both the selected candidate and the confidence signal: raw score rises to 31.67%, mean confidence drops to 0.396, and ECE falls to 0.093. The remaining OCER of 12.00% is still nonzero, so the result should be read as mitigation rather than elimination.

Retrieval is the strongest cautionary baseline: B4 has 0.00% OCER and 30.33% raw score, only 1.33 points below Q3C. Q3C is therefore clearest against direct confidence, temperature scaling, generic learned self-evaluation, and sampling-only uncertainty; against retrieval it is a trade-off between higher raw candidate accuracy and overconfident-error exposure at a particular operating point.

Family-level results, tabulated in Appendix Table 2, show why aggregate metrics are insufficient. On Natural Questions, Q3C reaches 59.00% raw score and 57.00% effective score, improving over both B0 and B4. On TriviaQA, retrieval has the best raw and effective score at 51.00%, while Q3C's abstention rate of 79.00% suppresses many answers. On TruthfulQA, all methods have low raw correctness, and Q3C's 98.00% abstention rate removes the few correct answers from effective credit. The family breakdown therefore identifies where Q3C currently helps: naturally phrased factoid QA with enough candidate/evidence signal, not misconception-sensitive truthfulness under a conservative threshold.

6 Analysis and Ablations

Paired significance. Table 4 reports the paired binary tests used for paper claims. The raw-score and OCER gains over direct confidence are strongly supported on this task matrix. Effective score is not significantly better than B0 because Q3C abstains on many rows, and Q3C does not significantly beat retrieval on raw score. We therefore state paired-task evidence, not seed-robust or universal dominance.

Component evidence. The ablations in Table 3 show mixed component effects. Removing evidence lowers raw score by 1.33 points and raises OCER by 9.00 points relative to full Q3C, supporting evidence-conditioned candidate scoring. Removing alternatives and semantic features has smaller or mixed effects, and the fixed-threshold ablation exactly matches full Q3C on aggregate raw, effective, and OCER metrics. The generic self-evaluation adapter is also important: it has the same raw score as direct generation and a high OCER of 64.00%, indicating that simply training a model to evaluate itself is not enough under our input and threshold design.

Threshold sweep. The post-hoc threshold sweep over completed Q3C rows finds the best effective score at threshold 0.00 for every family: 31.67% overall, 59.00% on Natural Questions, 34.00% on TriviaQA, and 2.00% on TruthfulQA, all at 100% coverage. This is not a new method result because it was computed after inference. It does show that threshold selection is a major utility lever and that future work should pre-specify family-aware or cost-aware operating points.

Risk-coverage interpretation. The sweep separates the learned ranking signal from the deployment policy. At threshold 0.00, Q3C behaves like the no-abstention variant and keeps all selected answers; at the recorded operating point, it discards many selected answers. This mirrors selective-prediction work: risk reduction and coverage preservation must be reported together (Luo et al., 2026; Srinivasan et al., 2024). For Q3C, the open problem is not merely to lower confidence but to choose an operating point that preserves correct answers while suppressing confident errors.

Benchmark / source	Model / backend	Method	Budget / decoding	Raw	Eff.	OCER	ECE
TriviaQA + NQ + TruthfulQA; 300 paired tasks, 100 per family	Qwen2.5-7B-Instruct / vLLM	B0 direct confidence	1 greedy answer	22.00	22.00	76.00	0.679
		B1 temperature scaled	1 greedy answer	22.00	22.00	7.33	0.437
		B2 self-consistency	6 sampled candidates	19.67	19.67	8.33	0.305
		B3 semantic entropy	6 sampled candidates	19.67	19.67	2.00	0.197
		B4 retrieval-only RAG	1 evidence context	30.33	30.33	**0.00**	0.212
		B5 self-RAG critique	retrieval + critique	12.00	12.00	83.00	0.791
		B6 generic self-eval	LoRA judge, no Q3C fields	22.00	22.00	64.00	0.577
		Q3C full	Q3C judge + abstain gate	**31.67**	24.00	12.00	**0.093**

Table 1: Main cross-benchmark matrix over 300 paired tasks: Q3C raises raw score from 22.00% to 31.67% and lowers OCER from 76.00% to 12.00% versus B0 under the same Qwen2.5-7B/vLLM evaluation backbone, while retrieval-only RAG remains non-dominated with 0.00% OCER. Raw and effective columns are percentages; secondary Q3C ablations are in Appendix Table 3.

Calibration versus usefulness. ECE improves for Q3C compared with B0, B1, and B6, but ECE alone would overstate the method's practical readiness. A system can be well calibrated at low confidence while abstaining from useful answers. Conversely, a retrieval baseline can have low OCER by assigning cautious confidence even when raw score is close to Q3C. We therefore interpret calibration metrics as diagnostics, not final deployment criteria. The paper-facing claim is that candidate-level evidence conditioning improves a specific confidence/accuracy trade-off on the completed evaluation matrix; it is not that the current threshold is an optimal policy.

7 Failure Cases and Discussion

TruthfulQA is the clearest negative result. Q3C has 100 TruthfulQA rows, 2 raw-correct rows, 98 abstentions, and 2 accepted answers. Both raw-correct answers are abstained, so the effective TruthfulQA score is 0.00% despite nonzero raw correctness; the abstained correct cases include the watermelon-seed and Neil Armstrong quotation questions. At the same time, 96 incorrect TruthfulQA answers are abstained, showing that the gate suppresses many errors but is too conservative to preserve benchmark utility.

The accepted incorrect cases show why lowering confidence is not sufficient: Q3C accepts an incorrect Darth Vader quotation and an incorrect Barack Obama birthplace answer with high confidence. This pattern matches symbolic-trigger and truthfulness diagnostics, where quotations, origins, named entities, and folk beliefs can make a common candidate look plausible while remaining benchmark-wrong (Lamba et al., 2025b,a). The practical next step is a contradiction-focused, pre-specified family-aware policy that preserves correct TruthfulQA answers, not simply a lower global threshold.

8 Conclusion

Q3C shows that a small open-weight instruction model, equipped with an adapted candidate-confidence judge, can reduce confident hallucination relative to direct raw confidence in the evaluated open-domain QA matrix. On the three-family matrix, Q3C improves raw score by 9.67 points and reduces OCER by 64.00 points versus B0. The same evidence keeps the claim calibrated: retrieval-only RAG remains non-dominated, adaptive-threshold gains are unsupported, and TruthfulQA exposes a harmful abstention collapse. The paper's main takeaway is therefore that evidence-conditioned candidate confidence is a promising mechanism for small-model open-domain QA, but its operating point is part of the method rather than a detail.

Limitations

The evidence is a single completed evaluation matrix with paired-task tests, not repeated random seeds or independent reruns. All methods use one 7B-class backbone, so the result may not transfer to other model families or larger models. The benchmarks cover three real QA families but not long-form, multi-hop, or interactive retrieval settings. The threshold sweep is post-hoc analysis and should not be interpreted as a deployed policy. The B6 generic adapter is included as a baseline after a held-out diagnostic, but it is not a broad study of self-evaluation adaptation. Finally, the TruthfulQA failure means Q3C should not be deployed as a truthfulness-preserving abstention system without a repaired operating policy.

Ethical Considerations

Confidence and abstention systems can reduce harm from unsupported factual answers, but they also create access trade-offs when the model refuses answerable questions. The TruthfulQA failure illustrates this risk: a policy can suppress many wrong answers while also discarding correct ones. Any deployment should report both raw correctness and abstention-adjusted utility, audit subgroup/topic behavior, and expose uncertainty rather than presenting abstention as guaranteed safety.

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A Reproducibility Appendix

The appendix tables are grouped by the checks they support. Table 2 expands the aggregate result by benchmark family, showing that Q3C's raw-score gain is concentrated on Natural Questions, that retrieval remains the stronger TriviaQA answerer, and that TruthfulQA exposes the abstention collapse discussed in Section 7. These family rows are interpretive diagnostics rather than additional benchmark slices; each row uses the same 100 paired tasks per family as the main matrix.

Family	Method	Raw	Eff.	Conf.	OCER	Abst.
Natural Questions	B0	33.00	33.00	0.913	66.00	0.00
Natural Questions	B4	39.00	39.00	0.394	0.00	0.00
Natural Questions	Q3C	**59.00**	**57.00**	0.780	28.00	15.00
TriviaQA	B0	32.00	32.00	0.900	65.00	0.00
TriviaQA	B4	**51.00**	**51.00**	0.751	0.00	0.00
TriviaQA	Q3C	34.00	15.00	0.276	6.00	79.00
TruthfulQA	B0	1.00	1.00	0.884	97.00	0.00
TruthfulQA	B4	1.00	1.00	0.350	0.00	0.00
TruthfulQA	Q3C	**2.00**	0.00	0.133	2.00	98.00

Table 2: Family breakdown: Q3C is strongest on Natural Questions at 59.00% raw score, trails retrieval on TriviaQA, and has 2.00% raw but 0.00% effective TruthfulQA score under the current abstention gate.

Table 3 reports the component-removal checks behind the method claim. The main use of this table is negative as well as positive: evidence removal hurts the confidence/error trade-off, but the fixed-threshold row ties the full method on aggregate metrics, so the paper does not claim that the learned adaptive threshold is isolated by the current matrix.

Ablation removed or changed	Raw Δ	Eff. Δ	Conf. Δ	OCER Δ
No evidence	+1.33	0.00	-0.072	-9.00
No alternatives	0.00	+0.67	+0.020	+2.00
No semantic features	+0.67	+2.33	+0.026	+1.33
Answer-only input	+0.67	+2.00	-0.026	-5.33
Fixed threshold	0.00	0.00	0.000	0.00
No abstention	0.00	-7.67	-0.021	0.00

Table 3: Ablation deltas for full Q3C: evidence and semantic inputs help some metrics, but the fixed threshold ties the full method on aggregate metrics, blocking an adaptive-threshold superiority claim.

The supplementary package contains: (i) the completed raw matrix in JSONL format, (ii) generated TSV tables for aggregate, family, paired-significance, ablation, threshold-sweep, and failure-case analyses, (iii) run manifests identifying the evaluated backbone and method conditions, and (iv) the figure asset used for Figure 1. The paper-facing regeneration entry point rebuilds the result tables from the completed raw matrix and does not rerun inference. Evaluation uses the same 300 paired task identifiers for every method, with 100 tasks from each benchmark family.

Table 4 anchors the significance statements used in the body. The "Q3C-only" and "Other-only" columns count paired task disagreements for the named binary metric before applying the exact McNemar test, which is why the table supports a raw-score and OCER claim against direct confidence but not a universal dominance claim over retrieval.

Comparison	Metric	Q3C-only	Other-only	p
Q3C vs B0	raw score	38	9	0.00002490
Q3C vs B0	effective	32	26	0.51184231
Q3C vs B0	OCER	5	197	$< 10^{-8}$
Q3C vs B4	raw score	36	32	0.71630076
Q3C vs B4	effective	29	48	0.03953953
Q3C vs B6	OCER	10	166	$< 10^{-8}$

Table 4: Paired tests over 300 tasks: Q3C's raw-score and OCER claims are supported against direct confidence, while retrieval remains competitive and wins on effective score at this operating point.

The released replay interface is intentionally path-neutral: use the result-regeneration target supplied with the artifact bundle, then compare the regenerated aggregate table, paired-significance table, and failure-case table with the values reported in Tables 1–3. The required checks are row-count agreement, method-condition coverage, benchmark-family coverage, and equality of the headline raw-score, effective-score, OCER, and ECE values after rounding to the displayed precision.